

POKER MATH

**SIMPLE AND BASIC POKER MATH TO
HELP YOU CRUSH THE COMPETITION,
PILE UP MONEY AND FEEL LIKE A
PROFESSIONAL POKER PLAYER**



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Poker Math: Discover Simple and Basic Poker Math to Help You Crush the Competition, Pile Up Money, And Feel Like A Professional Poker Player

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Table of Contents

[Introduction](#)

[Chapter 1: Poker 101-The Fundamentals of Poker](#)

[Chapter 2: Basic Poker Strategies](#)

[Chapter 3: Counting The Number Of Outs](#)

[Chapter 4: Calculating Poker Odds](#)

[Chapter 5: Calculating Your Equity](#)

[Chapter 6: How to Determine Expected Value \(EV\)](#)

[Conclusion](#)

Introduction

I want to thank you and congratulate you for downloading the book, *“Poker Math: Simple and Basic Poker Math To Help You Crush the Competition, Pile Up Money And Feel Like A Professional Poker Player.”*

This book has actionable information on simple and basic poker math that can give you an edge over the competition, pile up money as well as help you feel like a professional poker player.

Generally, poker is considered a game of chance. In fact, many players normally rely on their sixth sense and lady luck to guide them through poker play-offs. Well, the truth is that in a game of poker, while luck and senses apply to some degree, you need a lot more than luck and a sixth sense to beat the competition. So, what is it that you need? Simple: you need to master poker math if you truly want to play it like the pros do.

Well, you can win a poker game without using math. However, knowing how to calculate basic poker math will give you an edge over any opponent, and make you a consistent game winner. Who doesn't want that?

Players who do not understand that poker is a game of numbers are extremely disadvantaged even when playing against weak opponents. As a beginner poker player, you may feel overwhelmed by the numbers flying in

from every direction, and think that winning a game of poker is too difficult to comprehend. While this statement may hold some truth to it, the good news is that you do not need to be a math genius to understand poker math. By using easy, proven methods, you can learn poker math and become a great poker player.

I wrote this book after months of trying to outshine my friends by simply following my gut and luck with no consistent success. I wanted to find a formula that works and that wouldn't be dependent on something beyond my control. That's when I discovered that math could actually be applied in poker to give me an edge over the competition.

In this book, we shall look at simple poker math tricks and tips guaranteed to help you calculate outs and odds. After we do this, we shall go ahead to outline other basic poker calculations that will not only help you play fundamentally sound poker but also earn you lots of cash.

Thanks again for downloading this book, I hope you enjoy it!

Chapter 1: Poker 101-The Fundamentals of Poker

Before we delve deeper into the exciting world of poker math, let us kick off by looking at the basics of every poker game.

Poker 101: The Fundamentals of Poker

Before you can become a great poker player, you need some familiarity with common poker terms and factors. Therefore, it is only ideal that we start at the very beginning

Common Poker Jargon

The most basic poker terms are:

- ***Ante:*** This is a small portion bet that a player puts into the pot to fund the hand at the beginning. Most poker games use “blinds” to get the pot funded.
- ***Blinds:*** These are forced bets that the players to the left of the dealer post. Traditionally there are 2 blinds, the small blind and the big blind. The small blind is the person right next to the dealer and the big blind is the person next to the small blind.
- ***Bet:*** Wagered money
- ***Call:*** Indicates that you are determined to match the amount of a newly placed wager.
- ***Raise:*** Indicates your desire to increase a bet.
- ***Check:*** Indicates your unwillingness to raise without a wager.
- ***Fold:*** When you surrender your cards and exit from playing the hand.
- ***Limper:*** The first player to call a bet

The above are common and basic poker terms. Other than possessing knowledge of these terms, while playing a game of poker, irrespective of your opponent, there are a few things you should consider; below are some of these pertinent considerations.

Basic Poker Game Considerations

Here are a few things you should consider irrespective of the type of poker game that you are playing (friendly or professional):

1. You must consider whether the winning hand of a particular game is a high hand or low hand.
2. You must consider whether the ace is ranked high or low
3. You must consider whether all cards qualify to be a wild card.
4. You also must consider if the total cash you are allowed to wager on a particular hand has a pot limit.
5. You may also want to consider whether the deck is a standard 52 cards deck or different (i.e. if some cards have been added or removed).

Note: This book presumes that you have a little knowledge of poker. That's why we won't go into full details of the very basic things like the names of cards and other overly basic stuff. That's why this chapter is quite short and covers only the points I thought would be important to this topic.

Chapter 2: Basic Poker Strategies

Having looked at basic poker terms, and a few pertinent considerations you should have in mind as you engage in any poker game, let us swiftly move on to basic poker strategies

1st Strategy: Track Pot Size

In poker, before a hand is dealt, you should know how much is in the pot in order to know the odds its offering.

Example 1: Assume you are playing a no-limit Hold'em and it is a nine-handed table with \$100-\$200 blinds. Before anything happens, the pot has \$300. Don't calculate the number just once. If a player raises it to \$500 and another one calls, you know that the pot now contains \$1300.

To calculate the maximum bet in pot-limit poker, a nifty trick you can use is multiplying the last bet by three then adding all other bets plus the original pot. This may sound insane but it works.

Example 2: If the pot has \$10 and another player bets \$3, let's say you call the last bet. The pot size becomes $\$10 + \$3 + \$3 = \16 . This totals your maximum raise. Next, call \$3; your maximum bet will total \$19. Now apply our trick; $3 \times 3(\text{last bet}) + \text{plus } 10(\text{original pot size})$ will equal \$19(maximum bet).

You can apply this trick even with the blinds on the table in the first round. Say a table has \$1 small blind and \$2 big blinds. The initial max bet will be $\$3 \times \$2 + \$1 = \7 . If there are 2 limpers in front of you, the max bet becomes $\$3 \times \$2 + \$1 + \$2 + \$2 = \11

Knowing the most accurate pot size is not important if you are a beginner poker player. The idea here is to know the starting pot size and estimate it frequently; with practice, you will become an expert at observing the pot size.

2nd Strategy: Know Your Stack Size

As a beginner poker player, you can easily forget your stack size. For example, when playing no-limit Hold'em, you may make a normal opening and pot-commit after another player has already re-raised. Before the hand starts, you should know what your stack dictates: in this case, open raising all in or not.

This also applies when someone raises before you- do you need to re-raise all in or re-raise at a lesser amount? As a preference, maintain chips at the standard size (in stacks of 20) to make them easy to count and estimate frequently. You can organize your chips in creative ways such as building castles or arranging them by different denominations.

3rd Strategy: Select Profitable Hands

To make poker profits, you first must select profitable hands. At the end of the day, there is not an exact method to calculate how many hands you should play-. A conservative player will play less than an aggressive player. Sounds obvious but it's the truth. No poker player is the same when it comes to the overall style of play. Use the principles and practices in the rest of this book and go find out what style works best for you. In general, most inexperienced poker players will play more hands than they probably need to play.

‘Starting hand selection’ is opting for a good hand instead of a bad hand, and consequently folding the bad hands when you come across them. The fact is some hands win more than others do whether you are playing Texas Hold'em, Omaha, 7 card studs, or any other type of poker. Therefore, if you stick with good hands, you will see better results in the end (We shall look at and learn more about hand ranking in the next chapter).

4th Strategy: Size Your Bet

Most of us know how to bet. However, usually, most of us do not place much thought on bet sizes. Many poker players fall victims to making incorrectly sized bets at the wrong time. You must carefully determine the size of your bets and not just mash the pot button or merely bet a random number.

When playing poker, precision is important; when your bet size is appropriate, you get an edge over your rival. When playing poker, you should have a goal in mind. Are you placing a value bet (wanting to be called), or trying to make that player fold?

Your ultimate goal affects your bet. Think of your goal for a moment before you hammer that bet-pot button. Mix up your play i.e. do not make the same bet in similar situations all the time, to avoid becoming an easy read. Vary your bet sizes with regard to your ultimate goal.

For example, if you want your opponent to fold, bet as little as possible. This does not mean you should start min betting every hand; if you do, you will look like a fish. The trick here is to save money and simultaneously accomplish your goal. If you raise pre-flop in position, are called, and you would like to make a continuation bet, there is no need to bet the whole pot. Often a half to two thirds pot bet will produce similar results as a larger bet while risking fewer chips. When it comes to making your opponent fold,

smaller bets are just as effective as larger bets. Therefore, when you bet small, you save lots of money when called.

Now that you know some basics, let us move on to basic poker math you need to know and how to calculate it.

Chapter 3: Counting The Number Of Outs

Before you can calculate poker odds and other calculations, you need to learn how to calculate outs. Outs are cards remaining in the deck that improve your hand and make it superior to what you assume your opponent is holding.

In short, 'outs' are the number of winning cards. When playing poker, your best-case scenario is putting your money on the best hand at all times. However, this is not always the reality; sometimes, you are going to put your money on a bad hand. In such worst-case scenarios, the idea is to make a hand that is better than your opponent's. To do this, you need to count the number of unknown cards in the deck that could make your hand better. That is where your ability to analyze 'outs' comes in handy.

Poker Hand Rankings

To get you on the board, check out this list of poker hand rankings from the best to worst:

1. **Royal flush:** This is ten to an ace straight with all the five cards having similar suits. In poker, all suits rank equally.
2. **Straight flush:** A straight flush is whatever straight containing all five cards of the same suit.
3. **Four of a kind:** Four cards of the same rank. When any two players (on the board) share an identical four of a kind, the higher fifth card (the ‘kicker’) determines who wins the pot.
4. **Full house:** Refers to three cards of the same face or number value plus any two cards of the same rank.
5. **Flush:** When any five cards that share similar suits are not consecutive. The highest card amongst the five establishes the rank of the flush.
6. **Straight:** Five successive cards of different suits with aces either counting as a high or low card.
7. **Three of a kind, or trips:** This refers to any three cards of the same face or number value.
8. **Two pair:** A two pair is a hand with two different pairs of the same rank.
9. **One pair:** Whatever two cards that have the same rank
10. **High card:** Any other hand not mentioned.

Now that you know more about ranking your hand, let us move on to learning more about how to calculate your outs in a game.

How to Calculate your Out: Important Considerations

Calculating your out is not as complicated as it may seem at first glance; with a little practice, it will become second nature. Remember a suit has 13 cards. Here are some things you should consider:

1: Don't Count Your Outs Twice

When you have a straight and a flush draw, there are 15 outs. You may wonder why the outs are 15 instead of 17 since eight outs make a straight, and nine outs make a flush ($8+9=17$). In the example from table 2 below, the $A\heartsuit$, and $9\heartsuit$ make a flush and a complete straight. These outs should not be double-counted. Therefore, for this draw, our total outs are 15 not 17.

2: Blockers

A blocker is a card that you hold in your hand that automatically decreases the chances of your opponent winning on a specific type of hand. If you know that a card you hold is a card your opponent needs to make a winning hand then it increases your basic knowledge on their odds and yours.

Blockers come into play a lot when there is a straight or straight draw on the table. Say the flop is $A\spadesuit, 2\heartsuit, 3\heartsuit$ and you have $4\clubsuit, 4\diamondsuit$. You would hold 2 of the 4 outs someone with a 5 in their hand would need. Your two 4s would “block” that potential hand from your opponent and decrease the outs he needs to win the hand.

This is a very general overview in regards to blockers. Much more information can be researched on utilizing all the outs of blockers to your advantage. Use this basic knowledge as a building block in helping you determine your opponent's outs and odds. For more in-depth coverage on blockers go to <http://www.partypoker.com/blog/beginners-guide-blockers.html>

Take a quick look at the table below; it should give you a few ideas of a few common outs for post-flop play.

Outs	You Hold	Hope to Make
2	A Pair	Three of a Kind
4	Two Pair	Full House
4	Inside Straight	Straight
8	Open-ended Straight	Straight
9	Four Flush	Flush
15	Straight & Flush Draw	Straight / Flush +

The next table is a list of more draws with examples including specific outs you can use to improve your hand.

Type of Draw	Hand	The Flop	Specific Outs	# Outs
Pocket Pair to Set	2♦ 2♣	Q♦ 4♥ 9♣	2♥ 2♠	2
One Overcard	A♣ 8♦	J♠ 5♣ 2♦	A♠ A♦ A♥	3
Inside Straight Draw	J♥ 9♠	Q♣ 8♦ 4♠	10♣ 10♥ 10♠ 10♦	4
Two Pair to Full House	K♥ Q♣	K♠ Q♦ 5♣	K♣ K♦ Q♥ Q♠	4
One Pair to Two Pair or Set	A♠ Q♦	A♦ 10♠ 3♣	A♥ A♣ Q♥ Q♠ Q♣	5
No Pair to Pair	9♠ 7♦	2♣ 3♦ J♠	9♣ 9♥ 9♦ 7♠ 7♥ 7♣	6
Two Overcards to Over Pair	A♦ J♥	10♠ 8♦ 2♣	A♠ A♣ A♥ J♠ J♦ J♣	6
Set to Full House / Four of a Kind	6♠ 6♦	6♣ 7♥ J♠	6♥ 7♠ 7♦ 7♣ J♦ J♣ J♥	7
Open Ended Straight Draw	9♠ 8♦	7♠ 10♥ 3♣	J♦ J♣ J♥ J♠ 6♠ 6♦ 6♥ 6♣	8
Flush Draw	K♣ J♣	A♣ 6♣ 8♦	2♣-5♣ 7♣-10♣ Q♣	9
Inside Straight & Two Overcards	A♠ K♦	Q♥ 10♠ 2♣	J X A♦ A♥ A♣ K♥ K♠ K♣	10
Inside Straight Draw & Flush	A♦ K♦	J♦ Q♣ 3♦	10 X 2♦ 4♦-9♦ Q♦	12
Open Straight and Flush Draw	K♥ Q♥	10♥ J♣ 4♥	9 X A X 2♥-3♥ 5♥-8♥ J♥	15

Here are more examples to help you better understand outs:

Example1: With a Flush Draw

You hold A♥, 3♥ and the flop is 7♥, 9♣, K♥. In some scenarios, your ace high might already be winning, but in reality, your hand is not very strong. At least not at the moment.

However, if there happens to be another heart on the turn or river, make a flush, and if there is no player with a full house or better, you will win the hand. (It is not a paired board if so far no opponent has a full house.)

A standard deck has 13 cards of each suit and four of every value. You are holding two cards, and the other two are on the board. Therefore, out of 13

cards, four have been dealt, meaning that you still have nine hearts left in the deck. This indicates you have nine cards, which make you a probable winner, i.e. 9 outs.

Example 2: With a Straight Draw

Let us assume you have J♠, 10♠ and the flop is 6♣, Q♥, K♥. Any nine or ace will complete your straight. The deck has four nines and four aces so you have eight outs. If there is one missing card needed to complete a straight, you are left with four outs. For instance, if you had A♥ and J♠ as hole cards and the flop was K♣, Q♥, 7♦, your outs would be 10♣, 10♠, 10♥, 10♦.

Example3: A Straight Draw and Over Cards (Higher Cards on Board)

Assume you hold K♥, J♥ on a 10♦, A♠, 2♣ board. One of the four queens left in the deck will make you a straight. However, if your opponent gets a middle pocket pair e.g. 9♥ and 9♣ then you have got extra outs because having any king or any jack gives you a higher pair. In this case, your outs increase to ten (three kings, four queens, and three jacks).

Example 4: A Set Against a Flush Draw

If you hold 7♥, 7♦ on a 2♠, 7♠, J♠ board, your hand is pretty strong. However, you may not be a definite winner if any opponent has a two ♠s in his hand.

However, you have the chance to improve your hand even more. You have seven cards left (a seven, three twos and three jacks), so you will probably

be a full house or better, or if the turn and river rank the same, you will have a full house.

Example 5: A Straight Draw and a Flush Draw

If you hold 6♥, 7♥ on a J♥, 4♥, 5♣ board, you have both a flush draw and an open-ended straight draw. Therefore, you have eight outs to make the straight, and nine outs to make the flush. However, considering you double-counted two cards (3♥ and 8♥), you should deduct them. You are then left with 15 outs.

Now that you are adept at counting your outs, next you are going to learn how to apply this knowledge in some basic poker calculations.

Chapter 4: Calculating Poker Odds

After knowing how many outs are available to you, you need to calculate your odds. Do not forget to include only “good outs”. To simplify it, pot odds means what is in the pot is enough for you to call a bet. Poker odds give you the probability of winning any hand. In short, poker odds express the probability of getting a good card over a bad card (winning odds), or the size of the pot in comparison to the bet size (pot odds). Understanding the pot odds concept helps you differentiate between good bets and bad bets.

You can use several ways to crunch poker numbers and establish your chances of winning a hand and taking the prize. Let us discuss pot odds.

Calculate the Odds of Hitting Your Card(s) or Winning Odds

This is where you calculate odds using the number of outs as the basis.

Example

You hold Q♥, 9♥ on a K♥, A♥, 7♠, 4♠ board and your opponent's likely hand is A♦, A♠. Your calculations will be 13 cards in each suit: 4 hearts = 9 hearts remain. Ignoring what is in your opponent's hand, you have 9 outs.

To win, you will need J♥, 6♥, 10♥, 5♥, 7♥, 3♥, 6♥, 4♥, 2♥.

For instance, let us use your 9 outs to calculate a Texas Hold'em pot odds.

Steps

1: Calculate the number of cards you cannot see in the deck. With 52 cards in the deck and two in your hand, 50 remain. In addition, 4 cards are already exposed from the flop and turn, so 46 cards remain in the deck. With 46 cards unknown and 9 outs, you have 9 winning cards and 46 unseen cards, meaning 9 cards will make you win and 46 cards will make you lose.

That is:

$$52-2-4=46$$

2: Split the unseen cards into winners and losers (46-9=37.) In this case, 37 cards will cause you to lose and 9 cards give you a win.

3: Determine your odds. You have 37 to 9 odds of getting the cards you need on the river. Simplify this to $37 \div 9 = 4.1$ and round off to 4. Therefore, you have 4-to-1 odds of getting the winning flush. This expressed in a ratio is 4:1, or as a percentage, 20% (i.e. $4+1=5$ then $100/5$).

This simply means that your probability of losing this pot is 4 times more than winning it.

Note: Odds can be expressed ‘for’ or ‘against’. When you have large odds against winning, you are on the “long shot” which means you are not likely to succeed/win. The higher your odds, the less your chances of winning; therefore, with 100:1 odds, you are not going to win.

Calculate the Pot Odds

You need to understand the odds offered by the pot. The big question, in this case, would be ‘do you call the bet?’

Having roughly 4:1 odds of winning this hand, deciding whether to call the opponent’s bet depends on the amount of cash in the pot. Does it mean that if there is a substantial amount of cash in the pot, you should go for it? No. What you should look for is the ratio of the money you stand to win compared to your opponent’s bet size.

Let us continue with our initial example. Let us assume the pot has \$90 and your opponent bets \$10. That means that there is \$100 on the table as winnings. To see the river card, match your opponent’s \$10 bet. It will cost you \$10 to see if the last card dealt will be one of the nine cards you need to win.

Step 1: Work out the money you would generally win from your actual poker odds.

4:1 (your winning odds) X \$10(your bet) =\$40(the cash you’d normally win)

Step 2: Calculate how much money you stand to win and the odds offered by the pot.

$\$90$ (money in the pot) $+\$10$ (your opponent's new bet) = $\$100$ (what you stand to win)

Calculate pot odds simply: $100/10=10$ so the odds for your money are 10 to 1. This means that if you match your opponent's bet and win, you will make $\$10$ for every $\$1$ bet. Therefore, your pot odds will be 10:1.

Step 3: Choose whether to call the bet or not. In this case, by betting $\$10$, you have 10:1 odds when in reality, you have a 4:1 chance of winning. It is like when a bookmaker gives you 10:1 odds on a horse that has 4:1 winning chance. So should you call this bet? Of course, and move with speed because such odds assures you a great payday.

Summary

4:1 -odds against you winning: chances that you will lose 4 hands for every win

10:1 -odds offered to call: the $\$10$ bet earns you $\$100$.

Therefore, if this particular situation happened 5 times during play, it would be:

$\$100$ (won from 1 flush)- $\$40$ (lost from 4 losing hands) = $\$60$ (profit from 5 hands)

Converting odds into percentages example: 3 to 1; $3+1=4$ and $100/4=25\%$

Converting a percentage into odds: example 25%; $100/20=4$ and $4-1=3$ gives 3-to-1 odds.

Note: Call when the prize exceeds the cost, but fold if the cost is above the money you expect to win.

Basic Implied Odds

Implied odds are in the same category as pot odds. Implied odds deal mostly with certain situations: helping you decide if it is worth it to call a raise on a drawing hand (for a straight or a flush). Using implied odds can help you roughly figure out how much you can win if you end up hitting your draw.

The general rule is that if you can expect to win more money if you make your draw, then you have good implied odds. If you don't expect to win much, then you have bad implied odds.

Implied odds can't be truly calculated with a mathematical equation. Implied odds are learned through playing lots of hands, understanding your opponents, and knowing your outs for drawing hands. However, seeing certain scenarios can give you an idea of good and bad implied odds.

Example 1:

You have a 4♥, 5♠ and the board is 3♦, 6♣, ♣Q. You have an open-ended straight draw. If your opponent bets into the hand you have good implied odds because you can potentially win more money from him since your hand is somewhat disguised.

Example 2:

You have a 9♠, K♥ and the board is 6♣, 7♦, 8♦. If a 10 or a 5 comes on the board you could be playing against someone else who has a straight. You would have bad implied odds.

Here are 3 very basic rules for implied odds:

1. The more your hand is hidden the better your implied odds are.
2. If you have good implied odds, you don't necessarily have to use pot odds for that situation.
3. If you have bad implied odds you should stick with the pot odds to determine whether to call, raise, or fold.

Although you can't use any math to calculate implied odds, you can calculate how much you would need to win in order to make a call profitable. The calculation is simple:

pot odds - odds of hitting your draw

Example:

Say you have a straight draw off the flop and your opponent bets \$15 in a \$15 dollar pot which means you have to bet \$15 to win \$30.

Odds of hitting the straight (8 outs): 4.9:1 (4.88 according to the chart below)

Pot odds: 2:1

Draw odds-Pot odds: 2.9:1

The implied odds ratio is 2.9:1. If we multiply 2.9 by \$15 (the bet)= 43.50.

This means that if you were to call the \$15 bet, you would have to get at least \$43.50 from your opponent for the rest of the hand.

Below is a bonus handy chart of odds and outs you can memorize without doing the math.

Outs	Flop to Turn		Turn to River		Turn and River	
	%	Odds	%	Odds	%	Odds
20	42.6%	1.35-1	43.5%	1.30-1	67.5%	0.48-1
19	40.4%	1.47-1	41.3%	1.42-1	65.0%	0.54-1
18	38.3%	1.61-1	39.1%	1.56-1	62.4%	0.60-1
17	36.2%	1.77-1	37.0%	1.71-1	59.8%	0.67-1
16	34.0%	1.94-1	34.8%	1.88-1	57.0%	0.75-1
15	31.9%	2.13-1	32.6%	2.07-1	54.1%	0.85-1
14	29.8%	2.36-1	30.4%	2.29-1	51.2%	0.95-1
13	27.7%	2.62-1	28.3%	2.54-1	48.1%	1.08-1
12	25.5%	2.92-1	26.1%	2.83-1	45.0%	1.22-1
11	23.4%	3.27-1	23.9%	3.18-1	41.7%	1.40-1
10	21.3%	3.70-1	21.7%	3.60-1	38.4%	1.60-1
9	19.1%	4.22-1	19.6%	4.11-1	35.0%	1.86-1
8	17.0%	4.88-1	17.4%	4.75-1	31.5%	2.17-1
7	14.9%	5.71-1	15.2%	5.57-1	27.8%	2.60-1
6	12.8%	6.83-1	13.0%	6.67-1	24.1%	3.15-1
5	10.6%	8.40-1	10.9%	8.20-1	20.3%	3.93-1
4	8.5%	10.75-1	8.7%	10.50-1	16.5%	5.06-1
3	6.4%	14.67-1	6.5%	14.33-1	12.5%	7.00-1
2	4.3%	22.50-1	4.3%	22.00-1	8.4%	10.90-1
1	2.1%	46.00-1	2.2%	45.00-1	4.3%	22.26-1

Now that you know the theory and math behind calculating pot odds, in the next section, we will discuss a handy shortcut to figuring out your equity in

a pot.

Chapter 5: Calculating Your Equity

Your ability to calculate pot odds is half the journey to becoming a rich poker pro. After you are well familiar with pot odds, you need to know how to calculate your equity in a pot; pot equity is simply the share of the pot you have.

In short, your equity is the chances you have of winning the pot in comparison to your opponent. For example, if the pot had \$100, and your winning chance is 15%, your equity is \$15(i.e. 15% of \$100).

In poker, the basic principle of playing a drawing hand is that a pot needs to be large enough to call. When you have a straight draw, a flush draw, you will make your hand on the next card dealt almost 20% of the time (17% on a straight draw, 19% on a flush draw).

That is to say, 20% of the time, you will make your preferred hand, and 80% of the time you will not. The ratio 20 to 80 can be expressed as odds of 4 to 1. Thus, if the pot is 80 dollars, we can probably call a 20 dollars bet to break even. We use 'almost' because we have to make a profit, so, a call of \$20 is appropriate when there is more than \$80 in the pot.

Hand Equity: Hand equity generally involves counting your outs. Remember, if you are unable to tell what is in your opponent's hand, it is

better to assume they have the most dangerous hand compared to yours. If it is a flush draw, assume the draw is in their hand. If it is a paired board, imagine they have the full house or trips, if you are lucky. Folding your hand wrongly is less expensive than calling off your entire sack wrongly.

The easiest technique you can use to calculate your equity is to remember the following simple rule:

The Rule of Four and Two

The four and two method of calculating odds or equity is easy because you do not need a lot of precision, and if you are an Internet poker player, it's an easy way to calculate your winning chances. Using this method requires you to multiply your outs by four when you expect to see both the turn and river off the flop. However, when you have one card to go (i.e. turn to the river) you should multiply your outs by 2. You will get your chance of winning the pot as a percentage. Simply put:

- Multiply your outs by four on the flop
- Multiply your outs by two on the turn

Examples

For an open-ended straight draw (8 outs), your chances of making your straight with the two cards you expect to appear are 32% (8×4)

What about when you have just one card to come? Well, this is easier. Using a flush draw with 9 outs, it would be $9 \times 2 = 18\%$. For a straight draw, it is $8 \times 2 = 16\%$ which is 17%.

This method gives reasonable percentages for poker players who dislike mental gymnastics. However, in some cases, this shortcut may give slightly incorrect answers. For instance, in hands on the flop, with too many outs (> 8), to get a slightly more accurate figure, use this simple formula:

$(\text{Number of outs} \times 4) - (\text{number of outs} - 8) = \text{Equity}$

This means that when it comes to the equity of an open-ended straight flush draw, your calculations will be as follows:

$$(15 \times 4) - (15 - 8) = 53\%$$

If you do not use this formula in such instances, the percentage you get will be higher by seven points, which is a larger difference. If you make wrong equity calculations, you will not make sound poker decisions. Now let's put the two together:

Compare Your Odds to Equity

Now that you know your equity, and can calculate odds, how can you use these two aspects to tell if you have a good call or not? The first thing you need to do is convert the equity percentage to a ratio.

For example, with a flush draw, you get 9 outs on the flop. Therefore, your chance of making the best hand is $9 \times 4 = 36\%$. Remember pot odds are in ratio form, so to make comparisons between pot odds and your equity you need to convert:

How to Make Comparisons between Pot Odds and Your Equity

Step 1: convert that percentage into a ratio:

So 36% is described as 36 out the total 100. Therefore, $100-36=64\%$. Since you are calculating your odds and not your opponent's, your equity ratio is 64-36. To simplify this, use the following technique: always make the right side equal to one to simplify your ratio.

Step 2: Now, divide 36 by itself: $36/36=1$

Basic math rules state “whatever you do to one side of a ratio, you must do to the other” so, since you divided the right by 36...

Step 3: Also divide the left by 36: $64/36=1.77$. Your final ratio will be 1.7-1, meaning your chances of winning the pot are 1.7 against 1; therefore, for each time you make your hand, 1.7 times you do not.

Relax! You do not have to be that precise in your calculations. A simple short-cut you can apply is to estimate that 36 goes into 64 a bit less than twice. Therefore, it does not really matter if your answer is 1.6, 1.7, 1.8, or 1.9-1. Alternatively, round it off to 2-1, as long as it is close enough for you to decide whether to make the call.

So, how do you know if you should make the call based on the pot odds and the equity concept? Simply compare the numbers on the left of the two ratios (equity and pot odds).

Another basic poker principle is if you want to make money, your pot odds must be higher than your equity ratio. If your pot odds number is bigger than your equity, the call you are making is good, but if it is lower, then the call is bad.

For example, using our earlier example in the pot odds section, the odds were 4-to-1. To calculate your equity ratio, follow the lessons you have learned in this section as shown below.

$9 \times 4 = 36\%$. Therefore, your equity is 1.7-1, which you can simplify to whatever is close enough. This means your pot ratio (4-1) is more than your equity ratio (1.7-1). Your call could not be better.

Use the lessons you have learned about pot odds and equity to decide whether to call or not among other important decisions on any poker table.

Chapter 6: How to Determine Expected Value (EV)

You will find this term in many poker strategy discussions, thus, it is good to familiarize yourself with it.

Expected value is the speculative value of an uncertain outcome. It is like a prediction of the average. To calculate expected value, take each possible outcome, and then multiply each by the probability of it happening. Then add those numbers together.

This may sound tricky but you will get it eventually.

As you play poker, your main dilemma is making a choice with the highest expected value. For stringency sake, sometimes the highest expected value may be negative, but it is a less negative action.

For example, if you flip a coin, and expect to lose \$10 or win \$20, on average, you expect to win \$5 on each flip. Your EV is \$5.

How does expected value relate to poker?

Let's us talk about a very common scenario:

When playing Texas Hold'em, you hold K♥, J♣; the board has A♣, 10♣, 5♦, 8♣, 3♣ on a heads-up on the river situation. The pot has \$100, the big bet is \$10, and you are in the first position.

For argument's sake, let us say your opponent could be holding any two cards and could fold if he lacks a club. Let us also assume that your opponent will call a bet with any club and check with no clubs.

Let's do the math:

Given that your opponent could be holding any two cards, he, or she could as well be holding each of the individual clubs. However, for now, let us assume that your opponent does not have two of the cards because apparently, we know for sure that if our opponent had them, he or she would have raised the turn.

Note: In these scenarios, we will not bother adding in the times when the opponent does not have any clubs. In such cases, your opponent will fold if you bet, or check if you check. Therefore, you end up winning uncontested. Yes, it can affect expected value for the entire play, but not for the specific purpose in discussion here.

Scenario 1: You Bet!

If your opponent calls, you will certainly know that he or she will do so with an inferior hand. Otherwise, he or she would have bet a better hand. For your opponent to call, there are 6 possible cards he or she can use. Therefore, in those 6 times, you get an extra \$10. However, there are 8 cards available, so his or her chances of calling are 6/8 (six out of eight).
 $\$10 \times 6/8 = \7.5

If your opponent raises, you are aware that you have a worse hand, and you will lose \$10.

$$\$-10 \times 2/8 = \$-2.5$$

Therefore, in this situation your expected value is $\$7.5 + (-) 2.5 = \5 . That is not bad.

Scenario 2: You Check, With The Intention Of Calling If Your Opponent Bets

(As we said, ignore all the instances where your opponent does not have any club)

If you call your opponent's bet, you'll win 6 times out of 8 and will lose 2 times out of 8.

$$\$10 \times 6/8 = \$7.5$$

$$\$-10 \times 2/8 = \$-2.5$$

Again, your expected value is still \$5 in this scenario.

Theoretically speaking, in this case, checking is as good as betting, supposing you check with the intention to raise if your opponent bets.

Scenario 3: You Check With the Intention to Raise If your

Opponent Bets

Let us assume that your opponent always re-raises with the nuts. Therefore, if your opponent has $K\clubsuit$, he or she will re-raise you and you will fold. To avoid a lot of confusion, let us imagine your opponent calls your raise with any other club.

If your opponent has $K\clubsuit$, you will have to fold and lose \$20:

$$\$-20 \times 1/8 = \$-2.5$$

If your opponent has the queen, you will experience a showdown, but still lose \$20:

$$\$-20 \times 1/8 = \$-2.5$$

If your opponent has any other club, you will win:

$$20 \times 6/8 = 15$$

$$\text{Sum: } \$15 - \$2.5 - \$2.5 = \$10.8$$

Wrapping Up On Expected Value

In such a hypothetical situation, $\$10.8 - \$5 = \$5.8$ rounds off to \$6. Therefore, your EV is \$6 higher once you check and raise than when you bet out. Therefore, to maximize your chances of winning, check in this situation, and raise if your opponent bets, always. You will be in a position to get more profit than checking and calling (betting out). In addition, you will be a long-term winner.

Note: In poker, as well as other games, if your expected value is above zero, you will generate a profit in the end.

Conclusion

Knowing what your opponent will do is a million dollar question; unless you own a mind reading device, you cannot be sure what is on your opponent's mind. However, the poker math we have discussed can give you an indication of how most poker players make their plays.

All forms of 'correct' poker draw their basis on calculation we have looked at or similar types of calculations. The numbers dictate most poker strategies. Learning poker math is not about knowing how to play, but how to review and analyze your decisions in a bid to strengthen your game, thus allowing you to become a consistent winner.

Thank you again for downloading this book!

I hope this book was able to help you to understand how to do simple poker math to strengthen your play and gain an edge over the competition.

The next step is to implement what you've learned in this book.

Remember: practice makes perfect. Good luck, and may the poker gods smile upon thee!

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